

REPORT

Title

Prolog Program to Implement Forward Chaining

Aim

To implement a **Forward Chaining inference system in Prolog** using facts and rules and derive new conclusions through repeated rule application.

Problem Statement

Develop a Prolog program that uses known facts and predefined rules to derive new facts using Forward Chaining. The program should continue applying applicable rules until the required conclusion is obtained.

Requirements

- SWI-Prolog
- Computer system
- Prolog source file (`exp13.pl`)

Procedure

1. Open **SWI-Prolog**.
2. Create a new file named `exp13.pl`.
3. Define the initial facts.
4. Define the inference rules.
5. Implement the Forward Chaining predicate.
6. Save and load the program.
7. Execute the required queries.
8. Observe the derived conclusions.

Program

```

% Initial facts

fact(bird(tweety)).
fact(has_wings(tweety)).
fact(bird(eagle)).
fact(has_wings(eagle)).

% Rules

rule([bird(X), has_wings(X)], can_fly(X)).
rule([can_fly(X)], animal_can_move(X)).

% Forward Chaining

forward_chain(Facts, Facts) :-
    \+ new_fact(Facts, _).

forward_chain(Facts, FinalFacts) :-
    new_fact(Facts, NewFact),
    forward_chain([NewFact|Facts], FinalFacts).

% Find a new fact using a rule

new_fact(Facts, Conclusion) :-
    rule(Conditions, Conclusion),
    all_true(Conditions, Facts),
    \+ member(Conclusion, Facts).

% Check whether all conditions are known

all_true([], _).

all_true([Condition|Rest], Facts) :-
    member(Condition, Facts),
    all_true(Rest, Facts).

% Query predicate

```

```
query(Fact) :-  
    fact(F),  
    forward_chain([F], Facts),  
    member(Fact, Facts).
```

Required Queries

Load the program:

```
?- [exp13].
```

Output:

```
true.
```

Check the initial fact:

```
?- fact(bird(tweety)).
```

Output:

```
true.
```

Check whether Tweety can fly:

```
?- query(can_fly(tweety)).
```

Output:

```
true.
```

Check whether Tweety can move:

```
?- query(animal_can_move(tweety)).
```

Output:

```
true.
```

Check a fact that cannot be derived:

```
?- query(can_fly(penguin)).
```

Output:

```
false.
```

Sample Output

```
?- [exp13].
```

```
true.
```

```
?- query(can_fly(tweety)).
```

```
true.
```

```
?- query(animal_can_move(tweety)).
```

```
true.
```

```
?- query(can_fly(penguin)).
```

```
false.
```

Conclusion

Thus, the **Forward Chaining algorithm was successfully implemented in Prolog**. The program starts with known facts, applies applicable rules, derives new facts, and uses the derived knowledge to obtain further conclusions.